# Comparative Analysis of Ensemble Learning Methods for Multi-Class Smartphone Addiction Classification Among Adolescents

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***Abstract*—The academic performance of adolescents is influenced by various behavioral factors, with sleep patterns and smartphone usage emerging as key contributors. This study investigates the predictive relationship between these factors and academic performance using machine learning models. Data from adolescents on their sleep duration, smartphone screen time, and app usage were analyzed using models such as Decision Tree, Random Forest, Support Vector Machine (SVM), and gradient boosting methods (LightGBM and XGBoost). The models were evaluated based on accuracy, F1-score, and Area Under the Curve (AUC). Among the models tested, LightGBM achieved the highest accuracy of 86.2%, followed by XGBoost at 85.1%, indicating that boosting methods excel at capturing the non-linear relationships between the features. The analysis revealed that key factors affecting academic performance include sleep duration, time spent on educational apps, and overall smartphone usage patterns. This study underscores the importance of regulating screen time and prioritizing sleep to enhance academic outcomes for adolescents**

***Index Terms*—Academic Performance, Machine Learning, Sleep Patterns, Smartphone Use, Decision Tree, Random Forest, Support Vector Machine (SVM), Gradient Boosting, LightGBM, XGBoost, Adolescent Behavior, Predictive Modeling.**

## I. Introduction

Adolescence is a critical developmental period characterized by significant physical, emotional, and cognitive changes. Among the various factors influencing adolescent development, sleep and smartphone use have emerged as pivotal elements affecting academic performance. Sufficient sleep is essential for cognitive functions such as memory consolidation, attention, and learning, all of which are crucial for academic success. However, many adolescents experience inadequate sleep, often due to delayed bedtimes and early school start times, leading to sleep deprivation and subsequent academic challenges [1]. Concurrently, the pervasive use of smartphones among adolescents has introduced new dynamics into their daily routines. While smartphones serve as valuable tools for communication and learning, excessive use, particularly before bedtime, can disrupt sleep patterns. The blue light emitted by screens inhibits melatonin production, delaying sleep onset and reducing sleep quality, which in turn affects cognitive performance and academic outcomes [2]. Additionally, studies have shown that adolescents who use their smartphones late at night are more likely to suffer from disrupted circadian rhythms, leading to lower sleep quality and increased sleep latency, which ultimately impairs academic performance [3]. In Bangladesh, the rapid adoption of smartphones and the increasing academic pressures on students have raised concerns about the potential impacts on sleep and academic performance. Despite the growing body of research in this area, there is a paucity of studies focusing on Bangladeshi adolescents, particularly those who may be at greater risk due to socioeconomic factors [4]. This highlights the need for further investigation into the interplay between these variables in specific contexts.

This study aims to bridge this gap by employing machine learning techniques to analyze the relationship between sleep patterns, smartphone use, and academic performance among Bangladeshi adolescents. By leveraging data-driven approaches, this research seeks to identify predictive models that can inform interventions aimed at improving sleep hygiene and optimizing smartphone use to enhance academic outcomes.

## II. Literature Review

Several studies have examined the effects of smartphone use and sleep patterns on academic performance in adolescents, highlighting the growing concern over mobile addiction and its impact on various aspects of adolescent life. In the context of understanding how smartphone dependency affects students' sleep and academic outcomes, machine learning models have been increasingly used to predict and understand this relationship.

A significant study by Giraldo-Jiménez et al. [5] focuses on smartphone dependency among university students in Cali, Colombia, using machine learning models to predict dependency risks. In their research, they employed several algorithms, including Logistic Regression (LR), Random Forest (RF), and Support Vector Machines (SVM), on a dataset of 1,228 students. Their findings indicated that the most significant risk factors for smartphone dependency were long daily usage (over six hours), posture while using the device, and musculoskeletal discomfort. The study revealed that SVM and Random Forest models achieved the highest accuracy, demonstrating the potential of machine learning to detect dependency based on self-reported data. The study's results also emphasized the importance of device-related factors,

including posture, in determining the severity of dependency. While their models showed high sensitivity, they also noted the low specificity, which suggests the need for future research that incorporates physiological data to improve prediction reliability.

A related study by Alotaibi et al. [6] further expands on the use of machine learning in detecting smartphone addiction. Their research involved 500 participants from Saudi Arabia, where they collected data on smartphone usage patterns and behavioral indicators of addiction. By employing algorithms such as Logistic Regression, Decision Trees, Random Forest, and SVM, they found that the Random Forest classifier achieved the highest accuracy of 93%, with key predictors including daily usage duration, late-night usage, and the interference of smartphone use with academic and social activities. These findings highlight the significant role of machine learning in early identification of smartphone addiction and its potential negative consequences on academic performance. The study concluded that using predictive models for smartphone addiction could serve as a foundation for targeted interventions aimed at mitigating the adverse effects of smartphone dependency on students' mental health and academic success.

The relationship between smartphone use, sleep disruption, and academic performance is also well-documented in literature. Chindamo et al. [9] explored how nighttime smartphone use affects sleep quality and school performance in adolescents. The study, which involved 1,501 high school students from Italy, found that heavy smartphone use before bed led to shorter sleep duration and increased daytime sleepiness. This disruption in sleep was associated with poorer academic outcomes, as students with more daytime sleepiness were more likely to report lower grades. The study reinforced the idea that excessive evening screen time can interfere with sleep quality, which, in turn, negatively affects cognitive function and school performance.

Pérez-Chada et al. [10] also examined the effects of screen time on sleep and academic performance among Argentinean adolescents. Their research, which surveyed over 4,000 teenagers, showed that different types of screen activities—especially video game and mobile phone use—were linked to shorter sleep duration and increased daytime sleepiness. These sleep disruptions were found to significantly affect academic performance, with students who had inadequate sleep often reporting lower grades. This study highlights the need to address not only the quantity but also the timing of screen usage to improve both sleep quality and academic outcomes in adolescents.

In India, Mishra et al. [7] addressed the growing concern of mobile addiction among teenagers, particularly in the post-COVID-19 period, where screen time has increased significantly. Their study, which used both quantitative surveys and qualitative interviews, revealed that excessive mobile use negatively impacted sleep and academic performance. The researchers found that daily screen time, social media use, and sleep disturbances were the most significant predictors of mobile addiction. Using machine learning models such as Random Forest and Decision Trees, they demonstrated the predictive power of these models in identifying teenagers at risk of mobile addiction. This research underscores the need for early interventions to address the behavioral risk factors associated with mobile addiction, thereby reducing its detrimental effects on sleep and academic performance.

The increasing prevalence of smartphone use among adolescents has raised concerns about its impact on their overall quality of life. A systematic review conducted by Sohn et al. [11] examined the relationship between problematic smartphone use (PSU) and quality of life (QoL) among children and adolescents. The review, which included 15 studies from multiple countries, found a consistent negative correlation between PSU and QoL. Heavy smartphone use, particularly at night, was shown to displace face-to-face social interactions, reduce physical activity, and disrupt sleep, all contributing to a decline in emotional well-being and school performance. The review highlighted that emotional functioning, mood swings, and irritability were most affected, followed by school performance. These findings stress the importance of addressing smartphone addiction to improve not only adolescents' sleep quality but also their emotional and social functioning.

Additionally, a review by Fischer-Grote et al. [12] focused on the link between PSU and mental health outcomes, revealing that PSU was strongly associated with poor sleep quality and diminished school performance. Their systematic review of existing literature highlighted that smartphone addiction not only disrupts sleep but also affects other domains of quality of life, including social and emotional well-being. These findings support the notion that addressing smartphone addiction in adolescents is crucial for improving both their mental health and academic success.

In a conceptual review, Elhai et al. [13] examined the relationship between problematic smartphone use and mental health, specifically anxiety and depression. The study reviewed several theoretical frameworks, including the Displacement Hypothesis and Social Comparison Theory, to explain how excessive smartphone use can contribute to anxiety and depression. The review found moderate positive correlations between PSU and both anxiety and depression, suggesting that smartphone addiction may exacerbate mental health issues, further hindering academic performance. The study concluded that addressing smartphone addiction through targeted interventions can potentially mitigate these negative effects on both mental health and academic outcomes.

In conclusion, existing research has significantly advanced our understanding of the relationship between smartphone use, sleep disruption, and academic performance. Several studies have demonstrated that excessive smartphone use, particularly before bedtime, negatively impacts sleep, which in turn affects cognitive functions and school performance. Machine learning models have shown great promise in predicting smartphone addiction and identifying at-risk individuals. However, as highlighted by various studies, further research is needed to integrate objective data, such as real-time smartphone usage

logs and physiological measures, to improve the accuracy of predictions and better understand the causal mechanisms at play. Addressing smartphone addiction and promoting healthier screen time habits in adolescents could contribute to improved sleep quality, mental health, and academic success.

## III. Methodology

### A. Dataset Description

The dataset used in this project is the Teen Phone Addiction Dataset, which contains behavioral, psychological, and demographic attributes of adolescents. Key features include daily screen time, age, gender, self-esteem levels, depression levels, family communication, and academic performance. The target variable categorizes addiction severity into four classes: High, Moderate, Low, and Not Addicted. This dataset is well-suited to evaluate patterns of digital dependency and provides insights into risk factors contributing to problematic smartphone usage among teenagers.

### B. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was carried out to examine the behavioral, demographic, and psychological features of the dataset. This stage helped uncover patterns and correlations that guided preprocessing and model design.

*a) Grade Analysis:* The relationship between academic grade and smartphone addiction was explored to see if school level influenced dependency. Students in higher grades showed slightly greater usage, but the differences were modest. Addiction was found across all grades, suggesting that external factors such as peer influence and psychological traits play a stronger role than academic stage alone.

*b) Gender Analysis:* Gender-based analysis revealed broadly similar addiction patterns among male and female adolescents. While minor variations existed, high smartphone usage was prevalent in both groups, indicating that addiction is not strongly gender-dependent but rather widespread across adolescents.

*c) Result vs Gender:* Academic outcomes were compared by gender in relation to addiction. Both groups showed similar usage patterns, though heavy phone use occasionally coincided with weaker academic results. The effect, however, was inconsistent, implying that factors like study habits and motivation may moderate the impact of phone use on performance.

*d) Correlation Analysis:* Correlation coefficients showed that daily usage hours had the strongest positive correlation (0.60) with addiction. Sleep duration had a weak negative correlation (-0.22), while phone checks, apps used, time on social media, and gaming displayed weak positive correlations (0.25–0.32). Other features showed minimal or no linear relationship with addiction.

*e) Daily Screen Time Distribution:* Screen time analysis revealed clear differences across categories. Adolescents in the High addiction group consistently reported the longest daily usage, while those Not Addicted had the lowest. This confirmed screen time as a central behavioral marker of addiction severity.

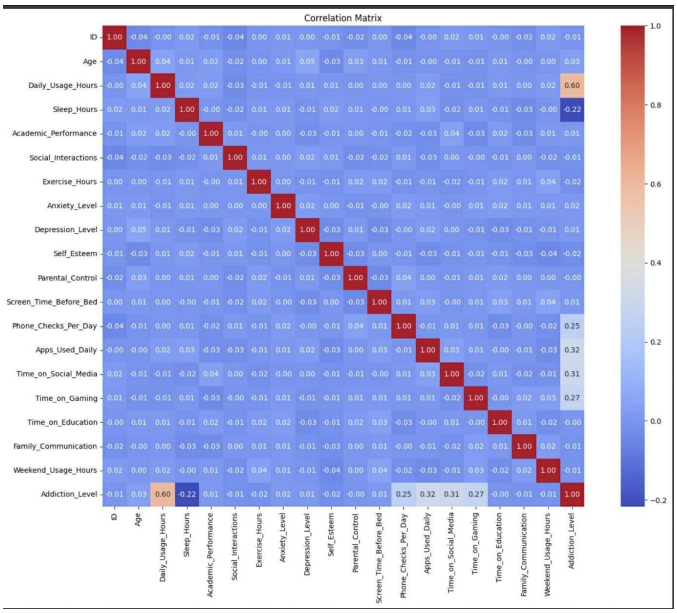



Fig. 1. Addiction Level vs. Behavioral and Academic Indicators

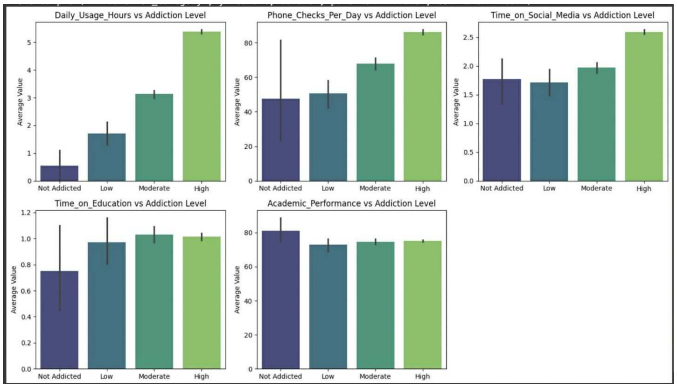



Fig. 2. Addiction Level vs. Behavioral and Academic Indicators

*f) Age-Based Behavioral Trends:* Behavioral trends across ages highlighted key patterns. Self-esteem remained stable, but depression peaked at age 18, likely linked to late-adolescent pressures. Family communication peaked around age 17 and then declined, while social media use consistently surpassed gaming and educational activities, reinforcing its dominant role in adolescent digital behavior.

*g) Family Communication vs Age:* When considered separately, family communication showed a peak at age 17 before declining. This reduction coincided with rising phone use, suggesting a possible trade-off between device time and family interaction.

### C. Data Preprocessing

To prepare the dataset for machine learning, the team implemented a series of preprocessing steps to enhance data quality. First, irrelevant columns such as ID, Name, and Location were removed, as they did not contribute to predicting addiction and could add unnecessary noise. Categorical features, like

gender, were converted into numerical values using label encoding to ensure compatibility with the models. Numerical attributes, such as daily screen time and age, were normalized to maintain a consistent scale and prevent larger values from skewing the results. To address the class imbalance, stratified sampling was used during data splitting to ensure proportional representation of each addiction category. Finally, exploratory data analysis revealed key predictors, including a moderate correlation between daily screen time and addiction, as well as the importance of depression and family communication. These steps ensured the data was clean and well-prepared for model training.

### *D. Learning Phase*

To classify the different levels of phone addiction, five machine learning algorithms were selected and systematically compared, each chosen for its unique strengths in handling classification problems. The first was XGBoost, a gradient boosting method well known for its ability to model complex, non-linear feature interactions and its robustness in dealing with imbalanced datasets, making it highly suitable for the uneven distribution of addiction categories in our data. K-Nearest Neighbors (KNN) was included as a simple, distance-based algorithm that classifies individuals by measuring their similarity to others in the dataset; while less sophisticated than boosting approaches, it offers a useful baseline to evaluate how well patterns can be captured purely from proximity-based relationships. CatBoost, another gradient boosting framework, was selected because of its efficient handling of categorical features and its ability to prevent overfitting by applying advanced regularization techniques. Similarly, LightGBM, a lightweight and highly efficient boosting algorithm, was employed for its speed and scalability, particularly useful for high-dimensional datasets with numerous features, as is the case in behavioral studies. Finally, Support Vector Machine (SVM) was used as a contrasting approach, relying on the construction of optimal hyperplanes to separate classes; it has a strong theoretical foundation in maximizing class margins and often performs well in binary and multi-class classification tasks with well-separated boundaries. Together, these models provided a diverse set of perspectives—ranging from ensemble-based learners to geometric and similarity-driven approaches—allowing for a more comprehensive evaluation of which techniques are most effective in predicting teen phone addiction. To ensure fairness in assessment, all models were evaluated using multiple performance metrics, including accuracy, precision, recall, F1-score, and the Area Under the Curve (AUC), measured across all four addiction categories.

## IV. Results and Discussion

### *A. Performance Metrics*

The evaluation of five machine learning models revealed clear differences in their ability to classify smartphone addiction levels. Among the models tested, the ensemble-based gradient boosting algorithms, LightGBM, XGBoost, and CatBoost, consistently outperformed the simpler distance- and margin-based methods, K-Nearest Neighbors (KNN) and Support Vector Machine (SVM). LightGBM achieved the highest overall accuracy of 86.2%, slightly surpassing XGBoost (85.1%) and CatBoost (84.5%). These findings suggest that boosting methods are particularly effective in capturing the non-linear and complex relationships in behavioral and psychological data. In contrast, KNN and SVM demonstrated lower accuracies of 78.8% and 79.0%, respectively, due to their limitations in handling overlapping classes and high-dimensional feature spaces.

A closer examination of class-specific metrics further highlights model performance. All boosting models achieved excellent results for the High addiction category, with F1-scores near 0.96 and AUC values above 0.95. This indicates that highly addicted adolescents show clear, distinguishable behavioral patterns, such as extended screen time and heightened psychological indicators, which the models could effectively learn. However, performance for the Low and Not Addicted categories was significantly weaker. Recall values for these minority classes were notably low across all models, resulting in F1-scores below 0.30. This suggests that simpler or less represented patterns in low-risk users remain difficult to capture.

### *B. Analysis*

Delving deeper into these results reveals several important insights. Notably, CatBoost outperformed other models for the Moderate addiction class, with an F1-score of 0.67. This indicates that CatBoost's capability to efficiently handle categorical features may allow it to detect nuanced behavioral differences among adolescents with moderate smartphone usage. On the other hand, KNN and SVM consistently underperformed across most categories, demonstrating their limited ability to manage imbalanced datasets and subtle inter-class distinctions. The findings underscore the advantages of ensemble-based models, particularly those designed to exploit feature interactions and handle categorical variables, in complex behavioral classification tasks.

Despite the strong performance of boosting algorithms, the low recall for underrepresented classes highlights an ongoing challenge: class imbalance. The subtle behavioral distinctions separating low- or non-addicted individuals from moderate users are not easily captured without additional data or advanced techniques to balance the dataset. Furthermore, the similarity between low-risk and moderate-risk behaviors increases the likelihood of misclassification, even for sophisticated models.

### *C. Implications*

The implications of these findings are twofold. First, gradient boosting models, particularly LightGBM and CatBoost, are highly effective tools for identifying adolescents at high risk of smartphone addiction, offering potential utility in early intervention programs. Second, the models' limitations in predicting low and non-addicted users suggest that additional strategies are necessary to achieve balanced performance

TABLE I
MODEL PERFORMANCE METRICS

| Model | Accuracy | F1-Score | | | | AUC (Avg) |
|---|---|---|---|---|---|---|
| | | High | Low | Moderate | Not Addicted | |
| XGBoost | 0.851 | 0.959 | 0.189 | 0.603 | 0.278 | 0.94 |
| KNN | 0.788 | 0.887 | 0.057 | 0.215 | 0.096 | 0.73 |
| CatBoost | 0.845 | 0.959 | 0.050 | 0.673 | 0.249 | 0.96 |
| LightGBM | 0.862 | 0.960 | 0.184 | 0.628 | 0.296 | 0.94 |
| SVM | 0.790 | 0.887 | 0.057 | 0.215 | 0.096 | 0.72 |

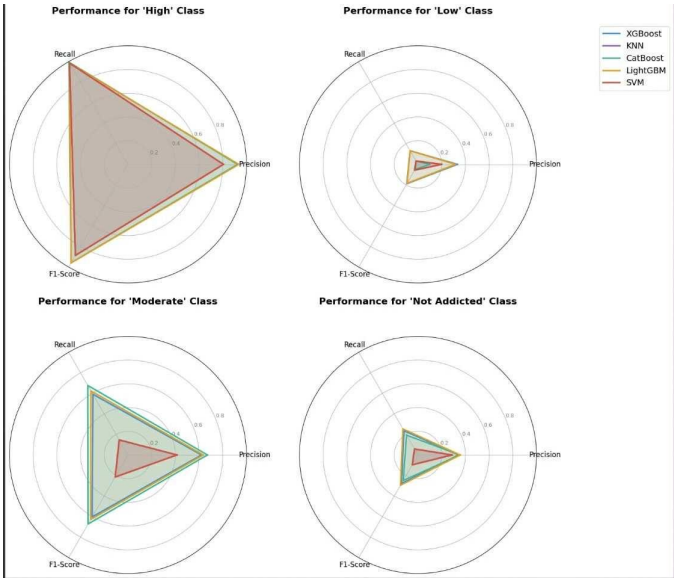



Fig. 3. Addiction Level vs. Behavioral and Academic Indicators

across all categories. Future work could focus on addressing class imbalance through oversampling techniques, such as SMOTE, or by expanding the dataset to include more low-risk participants. Incorporating additional contextual features, such as sleep patterns, parental monitoring, peer influence, and lifestyle habits, may further enhance model sensitivity to subtle behavioral differences. Ultimately, these improvements could lead to more comprehensive predictive tools capable of supporting targeted interventions to promote healthier smartphone usage among youth.

## V. CONCLUSION

This study assessed five machine learning models: XGBoost, KNN, CatBoost, LightGBM, and SVM, for classifying smartphone addiction levels in adolescents. Gradient boosting methods, particularly XGBoost and LightGBM, outperformed traditional models. However, predicting low and non-addicted categories was challenging due to class imbalance and subtle patterns. Using techniques like SMOTE and adding more contextual features could improve accuracy. The study highlights the potential of machine learning for early addiction detection, with future research focusing on addressing class imbalance and enriching the data for better predictive performance.

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